

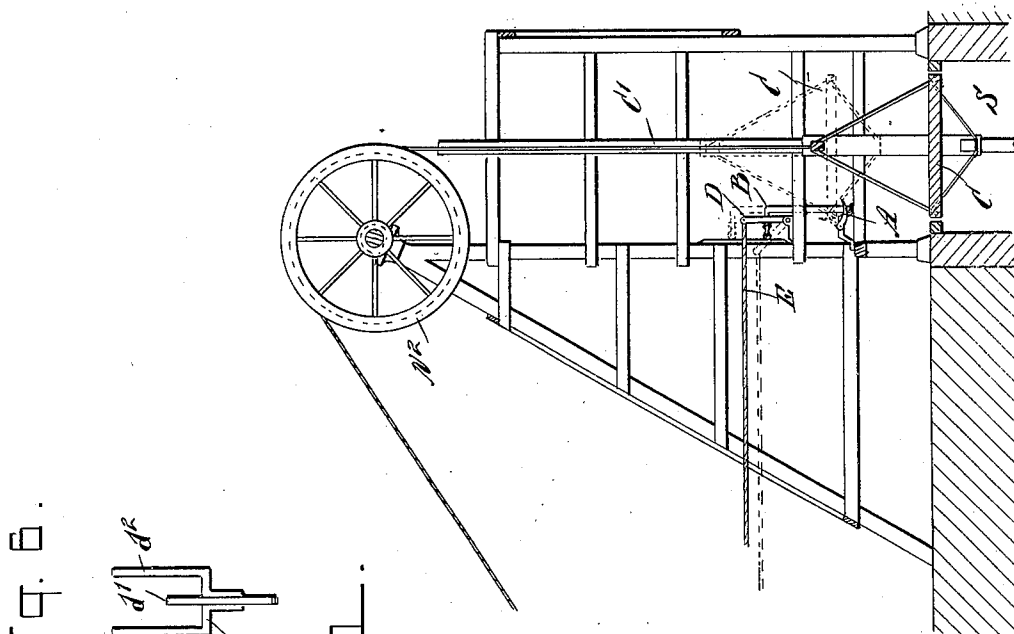
(No Model.)

2 Sheets—Sheet 1.

F. H. KOHLBRAKER.
AUTOMATIC STOP FOR HOISTS.

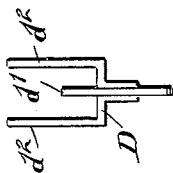
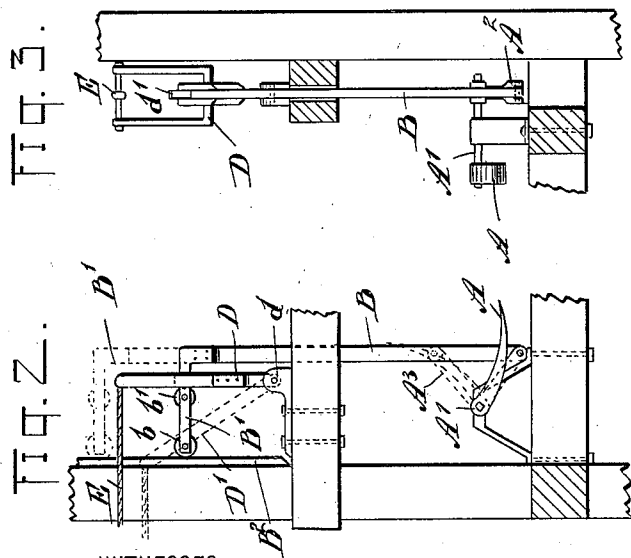
No. 601,515.

Patented Mar. 29, 1898.







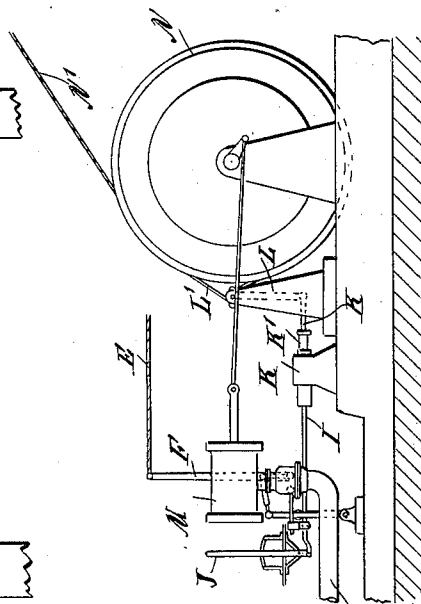
[illegible]

二五

三
 二
 一

WITNESSES :

H. Kellier.
H. L. Reynolds.



INVENTOR

F. H. Kohlbraker.

BY



ATTORNEYS.

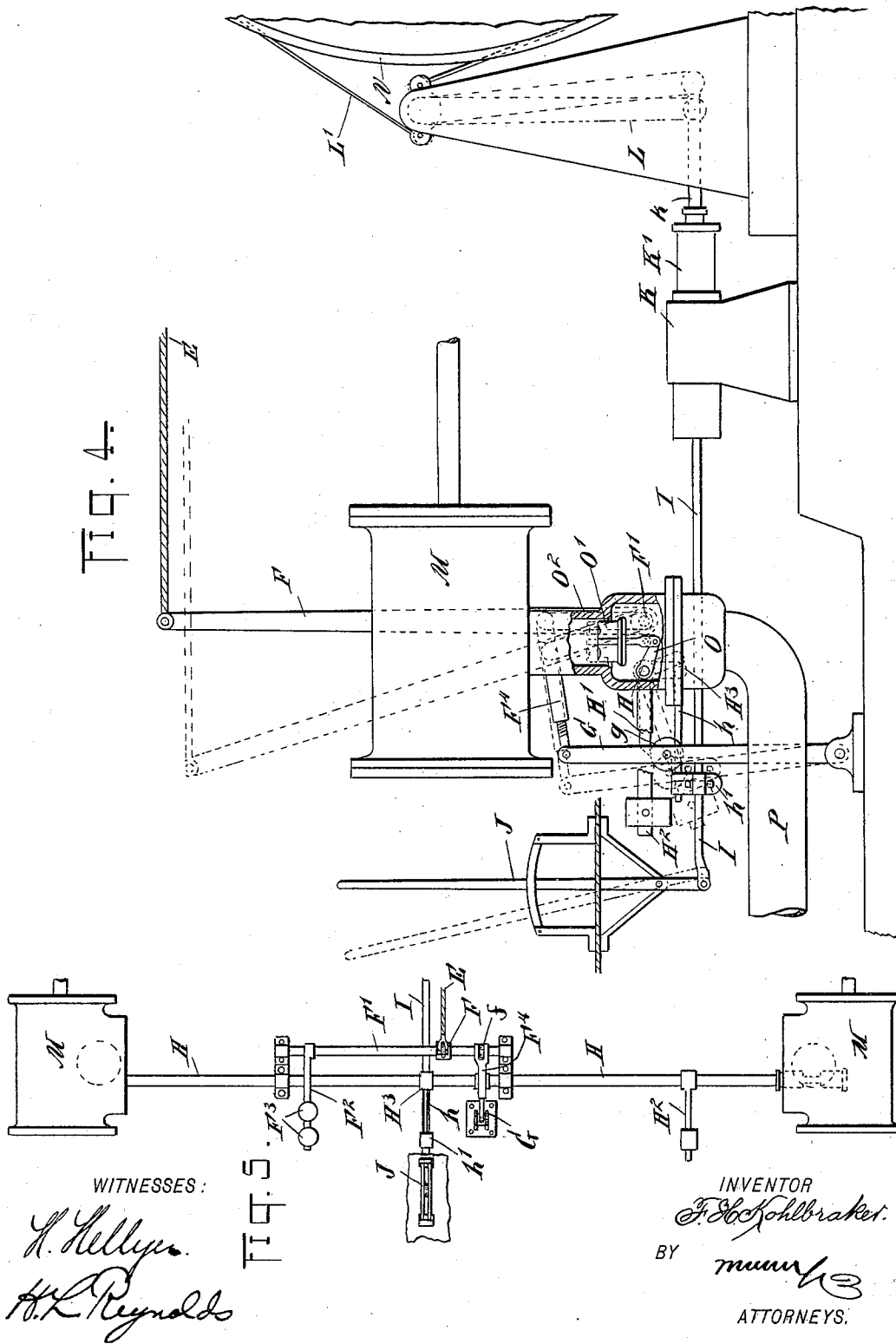
(No Model.)

2 Sheets—Sheet 2.

F. H. KOHLBRAKER.
AUTOMATIC STOP FOR HOISTS.

No. 601,515.

Patented Mar. 29, 1898.



UNITED STATES PATENT OFFICE.

FRANCIS H. KOHLBRAKER, OF SHAMOKIN, PENNSYLVANIA, ASSIGNOR OF
ONE-HALF TO MORRIS WILLIAMS, OF SAME PLACE.

AUTOMATIC STOP FOR HOISTS.

SPECIFICATION forming part of Letters Patent No. 601,515, dated March 29, 1898.

Application filed June 5, 1897. Serial No. 639,579. (No model.)

To all whom it may concern:

Be it known that I, FRANCIS H. KOHLBRAKER, of Shamokin, in the county of Northumberland and State of Pennsylvania, have invented a new and Improved Automatic Stop for Hoists, of which the following is a full, clear, and exact description.

My invention relates to an improved mechanism for stopping hoists or cages either in vertical shafts or slopes. Its object is to prevent their being hoisted beyond a certain point, and thus prevent the possibility of wrecking the cage or any part of the hoisting mechanism.

Reference is to be had to the accompanying drawings, forming a part of this specification, in which similar characters of reference indicate corresponding parts in all the views.

Figure 1 is a side elevation of my device. Figs. 2 and 3 are elevations of the lever-and-trip mechanism, which is directly engaged by the hoist. Fig. 4 is a side elevation with the valve in section, showing the steam-engine cylinder, the steam-brake, and the mechanism for controlling the same. Fig. 5 is a plan view of the shaft connecting the two steam-cylinders of the hoist-engine and the mechanism by which it is tripped, and Fig. 6 is a front elevation of the releasing-lever located at the hoist.

My invention is applicable to hoisting devices of any character using a steam-engine or similar motor for power. It may, in fact, be applied to water-wheels. I have herein shown it, however, as applied to an ordinary steam hoisting-engine and to a vertical shaft, although it may be as well applied to inclines.

The cage C is hoisted in the shaft S by any usual mechanism. I have herein shown the cable N' attached at one end to the cage and at the other winding upon a drum operated by a steam-engine and intermediate this passing over the sheave N². The shaft S is provided with vertical guides C', which control the position of the cage with considerable accuracy. Upon the framework or other suitable support adjacent to the side of the pit-shaft (see Figs. 2 and 3) is pivoted a shaft A', to which is attached by one end a lever A, said lever being placed in the path of the

hoist or cage. The shaft A' is provided with a second lever A² upon its opposite end, and this lever is connected to a releasing-bar B. This bar has a right-angle bend B' at its upper end and is provided with two rollers or pulleys b b'. One of these rollers engages a track or guide B². The other roller b' engages the central arm d' of the restraining-lever D.

The restraining-lever D is pivoted at its lower end by a pin d, attached to the framework or other suitable support adjacent to the shaft. At about the middle of its length the lever is forked, as shown in detail in Fig. 6, forming a yoke having two arms d². The central portion of this lever is continued upward a short distance beyond the yoke, forming the central arm d', which, as before stated, is engaged by the pulley b' and is held thereby against the action of the cord E, attached thereto. To the upper end of the arms d² is connected by any suitable means a cord or wire E, which is extended horizontally to a connection with the lever F of the tripping mechanism. This lever is mounted upon a horizontal shaft F', provided with a rearwardly-extending arm F², upon which the weights F³ are supported. The tendency of these weights is to keep the cord E taut, and thus keep the restraining-lever D held against the roller b'. In lieu of the weights F³ a spring may be employed. This is, however, recognized as an equivalent for a weight under such circumstances.

To the shaft F' is fixed a vertical bar f, to the upper end of which is connected a link F⁴, which is connected at its opposite end to the tripping-lever G, pivoted at its lower end to the floor or any suitable support. This tripping-lever G is provided with a roller g, normally supporting an arm H', fixed upon the horizontal shaft H, which is provided with a weighted lever H², acting to hold the arm H' down upon the roller g. The shaft H extends between the two steam-cylinders of the hoisting-engine and at each end enters a steam-chamber through which the steam for the engine passes. Within said steam-chamber the shaft H is supplied with a crank-arm O, to which is attached the throttle-valve O',

so that by a slight oscillation of the shaft H the throttle-valve O' may be closed and the steam entirely cut off from the engine. At some point intermediate the steam-cylinders the shaft H is also provided with a crank-arm H³, to which is attached a link h, provided with an adjustable clamp h', adapted to engage a rod I, extending to the steam-brake. The rod I is attached to a valve controlling the action of the steam-brake. A slight longitudinal motion of this valve is sufficient to admit steam to the brake and throw it in operation. This brake may be of any suitable form. As contemplated it consists of a steam-cylinder K', held in a suitable standard K and having one end of its rod k engaging one end of a lever L, forming part of a band-brake, the band L' of which embraces the periphery of the hoisting-drum M.

The rod I, controlling the steam-brake, may have a hand operating-lever J attached thereto, so that the steam-brake may be applied by hand, if desired. The steam-cylinder M of the hoisting-engine receives its steam through the chamber O², which is provided with the throttle-valve O', operated in the manner described.

The operation of my device is as follows: The cage as it rises from the shaft comes in contact with the lever A and through the connections of the rock-shaft A' raises the releasing-bar B until the roller b', carried thereby, is raised above the arm d' of the restraining-lever D. This frees the restraining-lever D, permitting it to swing to the position shown by dotted lines in Fig. 2 under the influence of the pull of the cord or wire E. As the lever F is connected by the wire or cord E to the restraining-lever D, the weighted arm or lever F² will rock the shaft F' enough to throw the roller g upon the tripping-lever G from beneath the end of the arm H', mounted upon the shaft H. As this shaft is provided with a weighted arm H², it will be rocked, so as to close the throttle-valve O', which is connected thereto, and at the same time to open the brake-valve, which is connected to the shaft by means of the crank H³, link h, and clamp h', secured to the brake-valve rod I. This opens the valve and applies the brake, thus stopping and holding the cage or hoist.

My device is not intended, ordinarily, to be relied upon for stopping the hoist at the proper point. This work is to be performed, as customary, by the engineer.

My device is intended as a safeguard to prevent injury should the engineer not be at his post at the proper time or neglect to shut off the steam sufficiently early. It is therefore located so that it will not be operated if the hoist is stopped at the proper point and is only operated when the hoist travels beyond its proper point. It may, if desired, be arranged so as to be operated at every hoist and relied upon to stop the engine at the

proper point. This, however, is not thought advisable.

Having thus described my invention, I claim as new and desire to secure by Letters Patent—

1. An automatic stop for hoisting-cages, comprising a pivoted lever projecting into the path of the cage, a releasing-bar mounted to reciprocate and actuated by the said lever, a valve-closing mechanism, and a restraining-lever connected with said valve-closing mechanism and normally engaged by the releasing-bar, substantially as described.

2. A hoist-operated valve-closing or stop mechanism, comprising a restraining-lever having one end formed as a yoke, and also having a central arm extending in the same direction between the arms of the yoke, a releasing-bar engaging said central arm, a valve-closing mechanism controlled by the restraining-lever, and a member connected with the releasing-bar and engageable by the hoist in its rise, substantially as described.

3. A hoist-operated valve-closing or stop mechanism, comprising a restraining-lever having one end formed as a yoke and also having a central arm extending in the same direction between the arms of the yoke, a releasing-bar having a pulley or roller attached thereto, and engaging the restraining-lever to hold the same, and a member connected to the releasing-bar and engageable by the hoist in its rise, to raise the releasing-bar and free the restraining-lever, substantially as described.

4. An automatic stop for hoisting-engines, comprising a power-operated brake, a weighted valve-operating lever connected with and operating the valves of the hoisting-engine and said brake, a tripping-lever normally holding the valve-operating lever out of action, a lever in the path of the hoist, and connections from said lever to the tripping-lever, whereby said tripping-lever is actuated to release the valve-operating lever, substantially as described.

5. An automatic stop mechanism for a hoisting apparatus, comprising a power-brake, an engine having a throttle-valve, a shaft having arms oppositely connected with the throttle-valve and the power-brake, a weighted lever and a supporting-arm connected to said shaft, a lever engaging said supporting-arm, an operating-lever in the path of the hoist, a weighted arm connected with said supporting-arm, and connections between said operating-lever and said weighted arm, for releasing said weighted arm and thereby to release the supporting-arm, substantially as described.

FRANCIS H. KOHLBRAKER.

Witnesses:

J. J. JOHN,

EDWARD BRENNAN.